

Blinkit End-to-End Analytics using Microsoft Fabric

1. Bronze → Silver → Gold Pipeline

Introduction

The Medallion Architecture is a modern data engineering design pattern used in Microsoft Fabric, Databricks, and Azure environments. It organizes data into multiple layers, where each layer improves the quality and usability of the data. In this project, the Medallion Architecture consists of three layers: Bronze, Silver, and Gold.

The primary objective of this architecture is to maintain raw data separately from processed data while creating optimized datasets for analytics, reporting, and machine learning.

Bronze Layer (Raw Data Layer)

Overview

The Bronze layer is the first stage of the data pipeline. It stores data exactly as it is received from the source system without performing significant transformations. This layer acts as the permanent storage for raw data and preserves all original records.

Tables in Bronze Layer

- `bronze_orders`
- `bronze_products`

- `bronze_stores`

Purpose

- Store raw transactional data.
- Preserve original records for auditing.
- Enable data recovery and reprocessing.
- Maintain historical data.

Operations Performed

- Data ingestion from source files.
- No joins.
- No aggregations.
- No business calculations.
- Minimal validation.

Example

The `bronze_orders` table contains:

- Order ID
- Customer ID
- Product ID
- Store ID
- Quantity
- Delivery Time
- Payment Method

These records are stored exactly as received from the source.

Silver Layer (Data Cleansing and Transformation)

Overview

The Silver layer transforms raw Bronze data into clean, structured, and analytics-ready data. This layer combines data from multiple tables and performs feature engineering.

Source Tables

- bronze_orders
- bronze_products
- bronze_stores

Transformations Performed

- Joining multiple tables.
- Removing invalid or null records.
- Calculating business metrics.
- Creating machine learning features.
- Standardizing data types.

Business Columns Created

Order Amount

$\text{order_amount} = \text{quantity} \times \text{price}$

This column calculates the total value of every order.

Order Hour

$\text{order_hour} = \text{HOUR}(\text{order_ts})$

Extracts the hour from the order timestamp to analyze hourly ordering patterns.

Late Delivery Indicator

is_late =

1 → delivery_minutes > promised_minutes

0 → otherwise

This becomes the target variable for the machine learning model.

Gold Layer (Business Layer)

Overview

The Gold layer contains aggregated, business-ready tables optimized for reporting and visualization. Instead of querying millions of transactional records, Power BI accesses these summarized tables.

Gold Tables

- gold_store_performance
 - gold_product_performance
 - gold_hourly_sales
 - gold_category_performance
 - gold_city_performance
 - gold_predictions
-

Gold Store Performance

This table summarizes the performance of every Blinkit store.

Metrics include:

- Total Orders
- Revenue
- Average Delivery Time

- Average Promised Time
 - Late Orders
 - On-Time Orders
 - Late Delivery Percentage
-

Gold Product Performance

Provides product-level business insights.

Metrics include:

- Orders
 - Total Quantity Sold
 - Revenue
 - Average Product Price
-

Gold Hourly Sales

Summarizes sales based on the order hour.

Metrics include:

- Orders per Hour
 - Revenue per Hour
 - Average Delivery Time
-

Gold Category Performance

Provides category-wise performance.

Metrics include:

- Total Orders
- Revenue

- Average Delivery Time
-

Gold City Performance

Provides city-level business insights.

Metrics include:

- Total Orders
 - Revenue
 - Average Delivery Time
 - Late Orders
-

Complete Pipeline Flow

Raw CSV Files

|



Bronze Layer

(Raw Data Storage)

|



Silver Layer

(Data Cleaning + Joins + Feature Engineering)

|



Machine Learning

(Predict Late Deliveries)

|



Gold Layer

(Business Aggregations)

|



Power BI Dashboard

2. Machine Learning Pipeline

Objective

The objective of the machine learning pipeline is to predict whether an order will be delivered late before the delivery occurs.

Problem Statement

Given the order details, predict whether:

- 0 → On-Time Delivery
- 1 → Late Delivery

This prediction enables Blinkit to proactively identify delayed orders and improve operational efficiency.

Target Variable

is_late

Definition:

$\text{delivery_minutes} > \text{promised_minutes}$

If true:

$\text{is_late} = 1$

Otherwise:

$\text{is_late} = 0$

Feature Engineering

The following features are used for prediction:

Numerical Features

- Quantity
- Price
- Promised Minutes
- Order Hour
- Peak Hour
- Day of Week
- Weekend Indicator

Categorical Features

- Store Name
 - Product Name
 - Category
 - City
 - Area
 - Payment Method
 - Dark Store ID
-

Feature Engineering Steps

Peak Hour

Orders placed during lunch and dinner hours are marked as peak-hour orders.

Peak Hour = 1

11 AM – 2 PM

6 PM – 10 PM

Otherwise:

Peak Hour = 0

Day of Week

The order timestamp is converted into a weekday value.

Example:

Monday = 2

Tuesday = 3

...

Sunday = 1

Weekend Feature

Orders placed on Saturday and Sunday are flagged.

Weekend = 1

Else = 0

Data Preprocessing

The following preprocessing steps are applied:

- Remove unnecessary ID columns.
- Handle missing values.
- Convert timestamps.
- Create engineered features.
- Encode categorical variables.
- Assemble features into a feature vector.

Encoding

Categorical columns are converted into numerical representations using:

StringIndexer

Converts text categories into integer indices.

Example:

Cash → 0

UPI → 1

Card → 2

OneHotEncoder

Transforms indexed categories into binary vectors for machine learning.

Feature Vector Creation

All numerical and encoded categorical features are combined into a single vector using:

VectorAssembler

Model Selection

The selected algorithm is:

Gradient Boosted Trees (GBTClassifier)

Reasons:

- High accuracy

- **Handles non-linear relationships**
 - **Performs well on structured datasets**
 - **Robust against overfitting**
-

Training Process

- 1. Read Silver table.**
 - 2. Apply feature engineering.**
 - 3. Encode categorical features.**
 - 4. Assemble feature vector.**
 - 5. Split data into training and testing datasets.**
 - 6. Train the GBT model.**
 - 7. Generate predictions.**
 - 8. Evaluate performance.**
 - 9. Save predictions into Gold layer.**
-

Model Evaluation Metrics

The model is evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

These metrics determine how well the model predicts late deliveries.

Machine Learning Workflow

Silver Data

|



Feature Engineering

|



StringIndexer

|



OneHotEncoder

|



VectorAssembler

|



Train-Test Split

|



GBT Classifier

|



Predictions

|



Evaluation



Gold Predictions Table

3. Power BI DAX Measures

Introduction

Data Analysis Expressions (DAX) are formulas used in Power BI to calculate business metrics. These measures provide dynamic calculations that update automatically based on filters and slicers.

Total Revenue

Purpose

Calculates the total sales revenue generated across all stores.

Total Revenue =

SUM(gold_store_performance[Revenue])

Total Orders

Purpose

Calculates the total number of completed orders.

Total Orders =

SUM(gold_store_performance[Total_Orders])

Average Delivery Time

Purpose

Calculates the average time required to deliver orders.

Average Delivery Time =

AVERAGE(gold_store_performance[Avg_Delivery_Time])

Late Delivery Percentage

Purpose

Shows the percentage of orders delivered after the promised delivery time.

Late Delivery % =

AVERAGE(gold_store_performance[Late_Percentage])

On-Time Delivery Percentage

Purpose

Displays the percentage of orders delivered within the promised time.

On-Time Delivery % =

AVERAGE(gold_store_performance[OnTime_Percentage])

Predicted Late Orders

Purpose

Counts the number of orders predicted by the machine learning model as late deliveries.

Predicted Late Orders =

**CALCULATE(
 COUNTROWS(gold_predictions),
 gold_predictions[prediction] = 1
)**

Dashboard KPIs

The Power BI dashboard includes the following KPI cards:

- **Total Revenue**
- **Total Orders**
- **Average Delivery Time**
- **Late Delivery Percentage**

- On-Time Delivery Percentage
 - Predicted Late Orders
-

Overall Project Workflow

Data Source



Bronze Layer (Raw Data)



Silver Layer (Data Cleaning & Feature Engineering)



Machine Learning (Late Delivery Prediction)



Gold Layer (Business Aggregation)



Power BI Dashboard (KPIs, Reports & Insights)